Paper Draft

Henry et al

LaTeX Help and Doc Standards

Example sentence that mentions Figure 7. Example sentence about some data (Fig. 7). Note how the refs are handled differently.

1 Introduction

Note to self: some of these (Vita?) citations aren't very good for what they're trying to accomplish; e.g. seems like they're citing research papers that included obligatory sentences, rather than seminal(ish) work those papers are citing. Will go through these then and double-check.

Aging is a complex biological process characterized by the progressive decline of physiological functions and increased susceptibility to disease, ultimately leading to death Shaposhnikov, 2022 159. Over the past several decades, research across diverse model organisms, from yeast and nematodes to flies and mammals, has revealed that the rate of aging is not fixed but can be modulated by genetic, environmental, and pharmacological interventions CITEShaposhnikov, 2022 159. Remarkably, many of the molecular pathways that regulate lifespan are highly conserved, suggesting that fundamental mechanisms of aging may be shared across the animal kingdom Miller, 2020 240CITE.

While much of aging research has involved direct perturbation of these conserved pathways, an intriguing dimension of aging biology was revealed by the discovery that sensory perception can modulate lifespan independent of the physical acts associated with perceived stimuli Pontillo, 2025 233Miller, 2020 240. Studies in the nematode worm, *Caenorhabitis elegans*, and the vinegar fly, *Drosophila melanogaster*, have demonstrated that sensory cues related to food, mates, and even the presence of dead conspecifics rapidly initiates physiological changes that influence patterns of aging Chakraborty, 2019 25Gendron, 2014 83[Sergei][Millie's paper]. These effects are mediated by neuronal circuits that emanate from sensory tissues and interface with deeper regions of the central nervous system, ultimately engaging conserved neuromodulators, including biogenic amines and neuropeptides, that influence metabolism, stress resistance, and longevity Miller, 2020 240Chakraborty, 2019 25.

In at least some cases, sensory perception has been shown to modulate lifespan through the motivational states *per se* it influences. For example, the neuropeptide F system, homologous to mammalian neuropeptide Y, is a component of reward-, and stress-related processing, and its activation can reduce starvation resistance and shorten lifespan Ryvkin, 2024 132Gendron, 2014 83. Sensory perception of potential mates shortens lifespan in male flies through circuits involving NPF signaling, whereas successful mating reverses these effects Gendron, 2014 83, suggesting that an unresolved expectation of reward can influence lifespan. Insatiable hunger also slows aging, despite promoting behaviors normally associated with decreased lifespan[1]. These observations indicate that identifying the neural circuits that interpret and integrate sensory inputs, as well as how they generate conserved internal states that prescribe physiological outcomes, may provide new insight into healthy aging across taxa, including in humans.

In *Drosophila*, one neural substrate has recently emerged as a strong candidate for such an interpretive center: the ellipsoid body (EB). The EB resides withing the central complex, which is a highly conserved region functionally analogous to the mammalian basal ganglia [strausfeld 2013]. It is subdivided into concentric domains innervated by X distinct classes of ring neurons (also called R-neurons) that process visual, thermal, mechanosensory, and circadian information Buhl, 2021 22Kahsai, 2012 138Yan, 2023 115[add more]. Exposure of flies to dead conspecifics, termed "death perception," reduces fat stores, lowers starvation resistance, and shortens lifespan in a manner that requires visual function and serotonin signaling through 5-HT2A-expressing R2 and R4 neurons Chakraborty, 2019 25. Insulin-like peptides, as well as the insulin-responsive transcription factor dFOXO, are required for aging effects of death perception, suggesting that EB ring neurons interface with conserved metabolic pathways implicated in lifespan control Gendron, 2023 128Pontillo, 2025 233. Additional roles for EB ring neuron classes in locomotor control, sleep, visual memory, and state-dependent action selection, suggest that it is well-positioned to link myriad sensory representations to organismal state Yan, 2023 115Kottler, 2019 266.

The extent to which the results in *Drosophila* may illuminate similar processes in other taxa may hinge on the nature of the EB's role in influencing health-related outcomes. One possibility is that these neurons function primarily as passive conduits or gates for lifespan-altering sensory information: if their function is disrupted, flies may simply fail to perceive[2] or internally represent salient stimuli, and phenotypes normally induced by those stimuli would not manifest. A more intriguing alternative is that the EB is an instructive modulator of aging-related processes, and direct manipulation of ring neuron activity, independent of any external sensory input, would be sufficient to alter aging. It is this latter case that, if true, suggests translational potential for targeting mechanisms of affective neuroscience REF in aging research.

Here, we use a variety of optogenetic tools, which enable temporally precise activation or inhibition of genetically-defined neuronal populations, to show that *Drosophila* EB ring neuron populations are presecriptive modulators of aging that encode behaviorally meaningful information and orchestrate diverse transcriptional and metabolic responses throughout the animal. Our findings indicate that these neurons can act as active regulators of pro-longevity physiology rather than only as conduits for perception-dependent aging signals. Activation of R2 neurons, inhibition of R4d neurons, and simultaneous activation of R2 and R4 neurons each extends lifespan in males and females, but these manipulations produce divergent transcriptional responses in head tissues, including distinct effects on insulin signaling, stress-response, and metabolic pathways. Despite these molecular differences, we detect no consistent changes in feeding, fat storage, locomotor activity, or climbing ability that account for the longevity effects, arguing against a simple explanation based on overt energetic or behavioral shifts. Valence assays further show that R2 activation alone produces a negatively-valenced state, whereas combined R2/R4 activation produces a positively-valenced state, indicating that distinct neural states with different immediate behavioral consequences can converge on increased lifespan. SOMETHING ABOUT BODIES AND METABOLOMICS HERE. Together, these findings support a model in which the ellipsoid body serves as an integrative neural locus through which sensory and state-related processes can influence aging via multiple downstream physiological programs.

2 Results

Manipulating subsets of EB neurons extends lifespan

To determine if the activity of ellipsoid body ring neurons is capable of modulating lifespan, we performed a series of experiments in which the activity of EB neurons was manipulated, with the following considerations informing our experimental design. First, we sought to eliminate potential developmental effects and to restrict neuronal manipulations to adults with a developed central complex[3]. Second, we wanted to avoid tools that require introducing confounding factors, such as high temperatures, which can significantly affect lifespan and obscure treatment effects on aging[4]. Third, we wanted the flexibility to perform acute, cyclic, or chronic manipulations in ethologically-relevant contexts that involve lifespan, behavior, and health assays. To accommodate these requirements, we adopted an optogenetic[5] approach to apply different treatment paradigms to freely-moving and fully developed adults by changing light-activation parameters and exposure times. To assay lifespans, we housed flies in custom optogenetic rigs with programmable, interchangeable LED bars, accomodating different channel rhodopsins and allowing us to either activate[6] or inhibit[7] targeted neuron populations. Flies were otherwise aged using standard husbandry conditions[8]. Unless otherwise noted, flies in all treatment paradigms were exposed to light-activation (40 Hz, 32 duty cycle) only during ZT 0-12 of a 12:12 light:dark cycle, beginning at 3 days post-eclosion and continuing throughout adulthood, promoting a normal circadian rhythm and restricting neuronal manipulations to periods when flies would typically be alert and interacting with their environment (fig:lifespanfigA).

[Table of lifespan reps]

We focused our investigation on R2 and R4, specifically R4d, neurons, as these are required for the perception of dead conspecifics to modulate lifespan[9] and for food perception to enhance starvation resistance[10]. When flies expressing UAS-CsChrimson[6] in R2 neurons (R2>Chr) were exposed to red light during subjective day, we observed a significantly extended lifespan in both females and males compared to siblings aged in the dark (Fig. 4B). This effect was not observed in genotype control flies that did not express CsChrimson, ruling out general effects of red light on lifespan and indicating a direct effect of neuronal activation. Activation of R4d neurons in a similar manner had no effect on lifespan (Fig. S1A). However, when R4d neurons were inhibited by exposing flies expressing the anion channelrhodpsin GtACR1[7] to green light during subjective day, lifespan was significantly extended in females and males (Fig. 4C). These results should be interpreted with caution, however, because we observed that exposure to high-intensity green light caused a significant reduction in lifespan in all cohorts (roughly 50% in females and 33% in males). Nevertheless, GtACR1 flies lived modestly but significantly longer than control flies in the light, and this difference was not observed in the dark.

The observation that R2 and R4d neuron activation had different effects on lifespan was somewhat unexpected given that measures of R2 and R4 neuronal activity were both observed to be higher following exposure to dead conspecifics[9], which accelerates aging[11]. We therefore decided to examine the effect of simultaneous activation of R2 and R4 (comprising R4d and R4m sub-populations) using a split-GAL4 driver. To our surprise, simultaneously activating both populations extended lifespan in males and females (Fig. 4D) to a greater extent than did R2 activation alone interaction p-value = XX. In this experiment, red light exposure modestly increased lifespan of CsChrimson-expressing flies as well as their genotypic controls, but the increase in lifespan was significantly greater in CsChrimson flies interaction p-value = X. Given the substantial increase in lifespan observed following simultaneous R2/R4 activation, we asked whether even broader manipulation of ellipsoid body neurons would recapitulate or even potentiate the lifespan extension that we observed following R2 and R2/R4 neuron activation. Using a driver that targets additional EB neuron populations, including at least R3 neurons, we observed that inhibition no effect on lifespan (Fig. 4E, left) and activation increased lifespan to an equal or lesser extent than R2 or R/R4 activation alone (Fig. 4E, right).

Together, these data demonstrate that enhanced EB neural activity promotes longevity primarily through R2 and R4 neurons, supporting the notion that the ellipsoid body is an instructive generator of lifespan-modulating signals. Reduced activity in the EB more generally has no apparent effect on aging in normal conditions. These results may be interpreted in the context of known activity patterns of R2/R4 in other contexts. Ishomoto et al. reported that R2, R4m, and R4d neurons work together in an incoherent type 1 feedforward loop (I1-FFL)[12]. Upon simultaneous activation of R2/R4m and R4d, R4d signaling initially increased, but sustained activity of GABAergic R2 and R4m neurons gradually suppressed R4d activity. If optogenetic activation recruits R2/R4 in this feedforward fashion, it may be that R2 activation, R2/R4 activation, and R4d inhibition are effectively the same intervention. Alternatively, they may engage partially or fully distinct pathways that converge on extended lifespan via different downstream mechanisms.

Different ring neuron manipulations produce distinct transcriptional profiles

To investigate how R2 and R4 neurons modulate aging and whether they do so via similar or distinct mechanisms, we performed bulk RNA-sequencing on brain-enriched tissues of female flies after one week of R2 activation, R4d inhibition, or R2/R4 activation, under the same conditions used in the lifespan assays described above. We chose to sample flies after one week of optogenetic manipulation to avoid survivorship bias that might arise at later time points while still allowing time for physiological differences to manifest. Flies were transferred directly from experimental rigs to liquid nitrogen, then decapitated by vortexing, and heads were collected with a sieve. Five biological replicates of 10 heads were collected for CsChrimson/GtACR1 and control genotypes from flies maintained either in constant darkness or channel-activating red/green lights during the 12 hours of subjective day. Whole head samples comprised primarily brain tissue, although other tissues were present, including the head fat body and chitinous exoskeleton tissue.

We used the DESeq2[13] package in R to identify transcriptional changes associated with enhanced R2 neural activity. We specified a statistical model in which genotype (CsChrimson, control) and light stimulation (yes, no) served as main effects. A genotype x light interaction term was used to identify significant transcriptional changes that were specific to light-driven channelrhodopsin stimulation, and these were the primary focus of our analysis. After one week of R2 neuron activation, we identified 123 genes (70 downregulated and 53 upregulated) whose expression values exhibited experiment-wise interaction p-values less than our cutoff of $p < 0.01$. Expression data from these genes produced clear separation of samples by treatment, indicating strong reproducibilty across each of the five replicates (Fig. 5A). Hierarchical clustering revealed that samples from flies with activated R2 neurons (i.e., R2>CsChrimson flies exposed to red light) formed a distinct group from flies without neuronal activation, including flies expressing CsChrimson but maintained in constant darkness as well as samples from the control genotypes maintained in either darkeness or red light.

Despite a stringent threshold of $p < 0.01$, we observed that one week of R2 neuron activation drove broad but pathway-enriched changes in the head. GO analysis of upregulated genes revealed enrichment for ribosome biogenesis and rRNA processing, insulin signaling, and response to starvation (Fig. 5B). Downregulated genes were enriched for protein folding and ER stress response pathways (Fig. 5C). Together, these results suggest that activation of R2 neurons induces a distinct metabolic state in the head, characterized by altered nutrient signaling and reduced ER stress. It also caught our attention that many of the differentially expressed genes have been implicated in aging biology, with several known to modulate aging directly. For example, we observed the upregulation of multiple genes consistent with dampened insulin-like signaling (e.g., *foxo*, *Thor* (encoding d4E-BP), *InR*, *dilp6*)[14]; upregulation of autophagy-related genes such as *Atg101*[15]; upregulation of *ImpL2*, an IIS antagonist which has been linked to lifespan extension[16]; and higher levels of *Cbs*, which encodes the rate-limiting enzyme in the transsulferation pathway and increases lifespan[17]. Together, these results demonstrate that short-term, part-time activation (12L/12D) of a few dozen R2 neurons drives significant changes in multiple aging-, nutrient-, and stress-response pathways.

Following one week of R4d silencing, far fewer genes were found to be differentially expressed (Fig. 5D), although R4d-specific changes were likley masked by remarkably potent independent effects of green light exposure. As observed for the R2 data, sample-based clustering grouped biological replicates as more closely related to each other than to samples from all other treatments, demonstrating reproducibility. Consistent with its significant negative effect on lifespan (Fig 1x), green light exposure affected the expression of 1,817 genes regardless of genotype (experiment-wise $p < 0.01$). However, only 11 genes (9 downregulated and 2 upregulated) were associated with a significant p-value for the interaction term of the model, which indicates a differential response in flies with R4d neurons silenced. Interestingly, in all treatments, green light drove strong upregulation of multiple genes that encode heat shock proteins (e.g., *Hsp23*, *Hsp26*, *Hsp27*, *Hsp70Bb*), which was significantly muted in flies with R4d neurons silenced. Other differentially expressed genes following R4d inhibition include the enzyme tyrosine hydroxylase (*ple*), which catalyzes the rate-determining step in dopamine synthesis[18],the serine protease homolog *scarface*[19], and several genes of unknown function.

This is probably a better discussion paragraph. Maybe move to there? Temperatures in our ventilated optogenetic lifespan apparatuses were not significantly higher than those experienced by flies aged in constant darkness, indicating that something other than heat was the primary cause of the effects of green light on life span and gene expression. Fly vision is more sensitive to green than to red light,[20] and the short lifespan (Fig. 4C) and over-representation of upregulated heatshock proteins suggest that it represents a meaningful stressor at the intensity used in our experiments and likely required for deep brain penetration. If so, it may be that a muted molecular response to moderate long-term stress contributed the extended lifespan of flies with silenced R4d neurons. It is, however, unknown whether similar mechanisms influence lifespan in more benign conditions.

We next asked how transcriptional changes following simultaneous activation of R2 and R4 compared to manipulations of the individual neuron populations. When R2 and R4 neurons were activated simultaneously, we once again observed excellent reproducibility with replicate samples within treatments being clustered together and flies with activated R2/R4 neurons forming a cluster distinct from both genotype and light controls (Fig. 5E). At a cutoff of $p < 0.01$, R2/R4 activation resulted in roughly twice as many transcriptional changes as did R2 activation alone (198 upregulated and 82 downregulated). A GO analysis of the upregulated genes was dominated by metabolic pathways related to amino acid metabolism as well as pentose-phosphate- and NADPH-generating processes (Fig. 5F). We also observed an upregulation of known lifespan-modulating genes involved in one carbon metabolism, including (*Cbs*, *Gnmt*, *Glyp*, *Glys*, *Nmdmc*, *Zw*). Genes involved in starvation responses and lipid and carbohydrate catabolism were significantly downregulated (Fig. 5G), which was in contrast to the upregulation of these pathways that we observed when R2 neurons were activated alone. Notably, rather than recapitulating the effects of either of the previous two manipulations, or impacting a set of genes that was effectively a union the effects of the focused manipulations, the transcriptional profile of R2/R4-activated flies was distinct from those of both R2 activation and R4d inhibition, even where it impinged on similar pathways (Fig. 5H-I).

EB ring neuron manipulations uncouple nutrient-state signaling from behavioral and physiological outputs

Manipulations of R2 and R4d neurons alone and in combination produced markedly divergent transcriptional profiles in head tissues (Fig. 5H-I). This is at odds with a strict reading of the I1-FFL circuit model[12], which would predict that these three interventions ultimately converge on a common downstream state. Our data are instead more consistent with functionally distinct activity patterns across R2 and R4 populations encoding different interpretations of sensory stimuli or internal physiological signals, in keeping with the ellipsoid body's established role as an integrative center. If each manipulation imposes a distinguishable internal state, those differences should in principle be reflected not only in gene expression but also in the behavioral and physiological programs those states normally coordinate. We therefore asked how R2 activation, R4d inhibition, and R2/R4 co-activation affected feeding, energy storage and mobilization, and organismal performance, readouts expected to track changes in nutrient state, motivational drive, and overall physiological condition.

Feeding behavior does not explain nutrient-state transcriptional signatures

mention that they were in light; so define "naive" better. Reduced nutrient intake, as well as homeostatic hunger drive, slow aging in *Drosophila*,[1,21], and GO analyses of differentially expressed genes in flies with activated R2 and R2/R4 neurons identified pathways indicative of altered nutrient state and metabolism. We therefore asked whether the effects of neuronal activation might include changes in feeding that indicate a hunger state or that underlie the observed transcriptional differences. To assay for feeding changes, we measured the volume of food intake using a consumption-excretion ("con-ex") assay[22]. Flies were placed on food containing a blue dye tracer for 24 hours, after which we quantified the amount of dye deposited on vial walls. Because flies can behaviorally and physiologically adapt to chronic neuron manipulation,[1], we tested light-naïve flies as well as flies that had been pre-treated with light for one week or one month, to evaluate changes over a significant portion of the lifespan.

update this with new stats We found no clear evidence that activation of R2 neurons alone or R2/R4 neurons in combination produces consistent changes in food consumption. With the exception of males of advanced age, R2 neuron activation resulted no significant effects on feeding. Naïve male and female R2>Chr flies excreted amounts of blue dye statistically indistinguishable from genetic controls (Fig. 6A, left panels for each sex). After a week of light pre-treatment, individuals from the same cohort of flies were randomly distributed among new vials and tested again (Fig. 6A, center). We again observed no light-driven changes in feeding by R2>Chr males or females. Following a month of light pre-treatment, males, but not females, exhibited a significant reduction correct direction? of feeding as evidenced by a statistically significant interaction effect (Fig. 6A, right). There were also no consistent effects of R2/R4 neuron activation on feeding. The interaction term was not statistically significant for naive males or females (Fig. 6B, left panels for each sex), while after a week of pre-treatment (Fig. 6B, center) we observed a modest increase in feeding in R2/R4-activated female (but not male) flies relative to genetic controls in red light (change: p interaction = 0.0044). After a month of pre-treatment (Fig. 6B, right), we observed no significant interaction between light and genotype in either sex. Not sure but maybe need a simple concluding sentence here.

I generally like this paragraph. Seems that it, too might be better in the discussion? I started making suggestions to this next paragraph, but got a bit lost in my own thoughts. We need to talk through this a bit I think Our GO analysis revealed markedly different transcriptional indicators of nutrient availablily in the brain following R2 and R2/R4d activation. General starvation-response processes were upregulated in R2-activated heads but downregulated in R2/R4d-activated heads ref to GO fig panels for each. Specific hunger-linked IIS genes, including *Thor* and *foxo*, were elevated after R2 activation, consistent with a starvation-like transcriptional program. One might predict such changes to result from actual nutrient scarcity if, for example, flies were consuming less. If, on the other hand, these transcriptional signatures reflected a hunger-like neural state, one might expect to observe increased feeding to satisfy that perceived demand. We observed neither, arguing that the transcriptomic changes are unlikely to arise secondarily from altered food consumption. In contrast, R2/R4d activation produced a transcriptome enriched for biosynthetic and storage-associated processes, including protein synthesis and glycogen metabolism, suggestive of a nutrient-replete state, yet again without the corresponding change in feeding that would be expected if this state were behaviorally driven. Together, these findings indicate that the transcriptional signatures elicited by these manipulations are almost surely a direct result of neural activation itself rather than a secondary effect of changes in behavior. More broadly, they argue against dietary restriction as the basis of lifespan extension in either manipulation and suggest that EB circuit activation can impose internal nutrient-state programs while uncoupling those states from the motivational and behavioral outputs that normally regulate feeding. Notably, this induced state also appears non-canonical: nutrient-responsive IIS changes were not accompanied by the coordinated physiological outputs normally associated with altered nutrient status, and reduced IIS in *Drosophila* is generally linked to reduced translation rather than to parallel engagement of anabolic processes[23]. Thus, direct EB neuron activation may differentially engage or uncouple nutrient-responsive signaling and biosynthetic programs, rather than simply recapitulating a normal starvation or satiety state.

TAG depletion does not predict starvation resistance after EB circuit manipulation

R2/R4 males are ALSO sensitized to starvation; think I mixed up the lines in Excel. Rewrite this. Male dark R2 is sig; so did they technically do worse? Different test? Okay. I skipped this until you update it. With adjusted p (BH), male 2769 isn't significant; definitely LOOKS different in the figure though

what about something like this and combine the two sections? If the starvation-like transcriptional signature induced by R2 activation reflects a true nutrient-deprived physiological state, we might expect flies to exhibit hallmarks of nutrient deprivation, including reduced triglyceride (TAG) stores as well as altered starvation resistance and activity levels.

Because nutrient-state-associated transcriptional changes following R2 and R2/R4d activation were not explained by altered feeding, we next asked whether these manipulations affected metabolic physiology more directly by measuring TAG stores and starvation resistance. After one week of R2 activation, TAG levels were unchanged in females but significantly reduced in males (Fig. 6C), whereas R2/R4 activation significantly reduced TAG in both sexes (Fig. 6D). We then tested whether these differences predicted survival under nutrient deprivation. Females of both genotypes were more sensitive to starvation than controls (Fig. 6E-F, left panels). In males, however, starvation outcomes diverged despite overlapping effects on TAG: R2>Chr males were indistinguishable from controls, whereas R2/R4d>Chr males were significantly more starvation resistant (Fig. 6E-F, right panels). Thus, reduced TAG did not consistently predict starvation sensitivity under these conditions. This dissociation was especially notable in R2>Chr females, in which longevity extension and starvation-associated transcriptional signatures occurred without detectable TAG depletion. Yet they were accompanied by reduced starvation resistance. Together, these findings indicate that EB neuron activation can alter lipid storage and starvation survival in ways not explained by food intake and not simply predicted by inferred nutrient state, reinforcing the conclusion that neural activation decouples nutrient-state signaling from canonical metabolic and physiological outputs.

We next asked whether these manipulations altered gross locomotor activity. We used the *Drosophila* Activity Monitoring System (DAMS)[24] to measure activity over 48 hours during optogenetic stimulation. Flies were housed individually in horizontal tubes and maintained either under activating light or in darkness for the duration of the assay. Acute R2 activation did not alter mean beam crossings in females (Fig. 1A, left; Light × Genotype, $p = 0.93$), but it modestly reduced activity in males (Light × Genotype, $p = 0.030$). [R2 1 week, Fig. 1A, center.] [R2 1 month, Fig. 1A, right.] [R2/R4 naive, Fig. 1B, left.] After one week of pre-treatment, R2/R4-activated flies of both sexes also exhibited reduced locomotor activity relative to dark controls (Fig. 1B, center; $p = 0.0052$ and $p < 0.0010$ for males and females, respectively), try not to shift measures mid sentence. with a significant interaction in females ($p = 0.035$) but not males ($p = 0.47$). Following one month of pre-treatment, this reduction persisted I don't hink it persisted in males if not there to begin with.. wait I got confused. I think I am reading mixed up between main effects and interaction. Just briefly mention main effects and focus everything about activation on interaction. in both sexes (Fig. 1B, right; p = 0.043 for females, p = 0.010 for males), although interaction effects were no longer significant (p = 0.079 and p = 0.12 for females and males, respectively) I don't think this sentence makes sense. Thus, neuronal activation did not increase gross locomotor activity under any condition tested. This result is particularly informative in R2>Chr males and R2/R4>Chr flies, in which TAG stores were reduced despite unchanged feeding and lower, rather than higher, movement. Moreover, we did not observe the hyperactivity classically associated with starvation-driven food-seeking in flies, consistent with our earlier finding that these manipulations impose nutrient-state-like programs without evoking canonical behavioral responses this continues a bit of my confusion with the paragraph above. Together, these data argue that increased gross locomotor activity is unlikely to account for the changes in TAG levels or starvation resistance caused by EB neuron activation.

Negative geotaxis reveals delayed, sex-specific effects of ring neuron manipulation

gtacr1 males are now the ONLY thing with sig ixn after correcting for multiple testing

Measures of baseline feeding, TAG storage, and spontaneous locomotor activity did not clearly distinguish the effects of R2 and R2/R4d manipulations, leading us to ask whether differences would manifest following an acute behavioral challenge. To assess this, we tested negative geotaxis, an evoked climbing response widely used to measure age-sensitive locomotor performance and neuromuscular function in *Drosophila*[25,26]. Using our custom "D-Drop" platform, which drops flies in individual lanes and uses video tracking to quantify several characteristics of their subsequent climbing behavior. We focused on two specific measures: maximum height reached 9 sec after drop and total upward distance climbed over 30 sec. R2>Chr flies of neither sex exhibited a significant Light × Genotype interaction for either metric after one week or one month of activation (Fig. 2A). R2/R4d activation likewise produced no detectable effect after one week of pre-treatment in either sex (Fig. 2B, left panel for each sex). After one month of pre-treatment, however, female R2/R4d>Chr flies outperformed controls in both height at 9 sec and total upward distance, whereas males showed reduced total upward distance and a similar trend toward reduced height at 9 sec (Fig. 2B, right panels). so no naive data for either? only a 1-hr point for R2; hard to do "naive" since there's no light treatment during the assay

Motivated in part by our transcription data that suggest R4d neuron inhibition may buffer stress responses to green light exposure, we asked if the effects of green light manifest as reduced negative geotaxies and if R4d inhibition would promote endurance against this stress. prob need to say something about green light in general first here...then about the second part In females, however, R4d>GtACR1 produced only a transient effect: after one week of pre-treatment, flies climbed less total upward distance than controls, without a corresponding difference in height at 9 sec, and this effect was no longer evident after prolonged inhibition (Fig. 2C). In males, by contrast, R4d inhibition improved both climbing metrics after one week pre-treatment, and this advantage persisted after five weeks of pre-treatment. Effects of R4d activation were transient by comparison (Fig. S4A). Transient how? Need to describe the data a bit. I should be able to get the jist without the figure Overall, after one week of pre-treatment, which corresponds with the time of our transcriptional analysis, negative geotaxis did not distinguish R2 from R2/R4d activation. Instead, behavioral differences emerged primarily at later time points and in a strongly sex-dependent manner. The most robust effect was the sustained improvement in male climbing caused by R4d inhibition. Although these later phenotypes may bear on the lifespan effects of these manipulations, the present data do not establish whether they arise as delayed consequences of the transcriptional programs observed after one week of treatment. not sure exactly what you are distinguishing here. Also, it might be more relvant to the discussion because it is not setting up for the next section. Could just stop with the previous sentence

maybe talk about this paragraph as well Overall, negative geotaxis did not clearly resolve the functional differences between R2 and R2/R4 manipulations. No effects were evident at the one-week time point corresponding to our transcriptional analysis, indicating that the early molecular states induced by these perturbations are not accompanied by parallel changes in evoked climbing behavior. Divergent effects emerged only later in R2/R4 flies, but whether these represent delayed consequences of the earlier transcriptional programs remains unclear. By contrast, R4d inhibition produced a robust and sustained improvement in male climbing performance. Because comparable transcriptomic data are not available for males, we cannot determine whether this sex-specific phenotype reflects a shared molecular program with different behavioral output or a distinct underlying mechanism. Taken together, these data indicate that our ring neuron manipulations do not significantly alter behavioral patterns and do not produce gross body weight or locomotor deficits. Transcriptional states characterized by GO do not map onto detectable differences in these common health and behavior readouts. Lifespan extension is therefore unlikely to be mediated by changes in these phenotypes, and may instead be due to direct effects of neuron manipulation on downstream signaling and metabolic pathways.

This can probably go: Based on our transcriptional profiles and on prior work showing that disruptions in IIS signaling can delay age-related decline in negative geotaxis[26], one might predict that prolonged R2 activation would improve climbing performance, as it induces downstream effectors of decreased insulin signaling (*Thor, foxo*).

R2, R4d, and R2/R4 activation produce distinct valence states

Activation of R2 neurons and co-activation of R2/R4 neurons produced distinct transcriptional responses without consistent effects on feeding, locomotor activity, or overt physiology, arguing against a model in which EB activity influences aging solely through a shared set of behavioral or physiological outputs. Rather, these results suggested that EB ring neurons may act instructively, imposing distinct internal states whose immediate consequences are separable but whose downstream effects converge on lifespan regulation. In natural environments, ring neuron activity is engaged by behaviorally relevant stimuli, including cues that drive discrimination of nutritive value[27] or avoidance of threatening conditions[28]. Because standard husbandry conditions are relatively benign and homogeneous, however, the motivational consequences of experimentally imposed EB states may be muted under baseline conditions. We therefore next asked whether these manipulations differ in valence, reasoning that their appetitive or aversive properties might clarify the functional logic by which EB circuits influence lifespan.

We first examined R2/R4 neuron valence using the FLIC assay, which permits flies to associate optogenetic stimulation with self-initiated interactions with liquid food at a defined well[29]. In this paradigm, individual flies were presented with a choice between a water well and an 10% sucrose well, and closed-loop optogenetic stimulation (red light for CsChrimson or green light for GtACR1) was triggered selectively by contact with the sucrose food. Each "trigger" fly was paired with a yoked control that received identical light stimulation determined solely by the behavior of the trigger fly, without relationship to its own feeding behavior (Fig. 3A). This design allowed us to ask whether activation or inhibition of specific EB neuron populations altered the motivational value of the sucrose-associated food while accounting for non-associative consequences of light exposure (increased hunger, for example). As with our behavioral assays, we performed these experiments both in light-naïve flies and in flies pre-exposed to light for one week or one month. Because FLIC data can be used to extract several features of feeding behavior, it allowed us to assess the valence of each manipulation using multiple behavioral dimensions. Each FLIC signal over threshold was scored as an "interaction" with the liquid food, and these data were further parsed into the number of discrete well visits ("events") and the duration of those visits ("median duration")[30]. Increased sucrose interactions, more frequent visits, or longer visit durations were interpreted as evidence of positive valence, whereas rapid abandonment of the trigger well or subsequent avoidance was taken to indicate negative valence. Due to the sophistication and limited throughput of this assay, and because our transcriptomic analyses were performed in females, we restricted this component of our study to females to enable direct comparison between observed valence and underlying transcriptional states.

Again, we first tested R2 activation, prompted by the observation that R2>Chr animals exhibited transcriptomic signatures consistent with nutrient stress and starvation-like state changes. If so, activation of R2 neurons might be negatively valenced or might alter the motivational salience of a nutritive stimulus, respectively. Indeed, light-naïve R2>Chr trigger flies showed a marked reduction in interactions with the sucrose well relative to yoked controls, indicating that contact with the trigger well was strongly aversive despite the inherent attractiveness of sucrose (Fig. 3B, left; $p < 0.01$). This effect was accompanied by a significant reduction in event duration (Fig. S2A, left; $p < 0.01$), suggesting that flies rapidly disengaged from the well once stimulation was triggered. Notably, however, R2>Chr flies did not reduce the number of visits to the sucrose well, implying that R2 activation did not simply condition avoidance of the well itself (Fig. S2A, right). Instead, these data suggest a more acute aversive state that promotes premature disengagement while leaving repeated approach behavior largely intact. This phenotype remained evident after one week (Fig. 3B, center; $p = 0.035$) and one month (Fig. 3B, right; $p < 0.01$) of light pre-treatment, indicating that the negative valence associated with R2 activation is persistent and does not substantially diminish with chronic exposure.

Because the transcriptional profile induced by R2 activation was associated with genes involved in starvation response, we considered whether the aversive effect of R2 stimulation might obscure a parallel increase in motivation to feed. In principle, if R2 activation enhanced food drive while simultaneously imposing negative valence, then the immediate aversive response of trigger flies could preclude increased sucrose consumption. To test this possibility, we modified the yoked design to include a single, wild-type trigger fly and two yoked animals, one R2>Chr and one genetic control (Fig. S2B). Under this configuration, both yoked animals experienced light delivery uncoupled from the food and at a frequency determined by normal feeding of the wild-type trigger fly. If R2 neuron activation increased hunger, we would expect the yoked R2>Chr flies to approach or interact with the sucrose well more often than yoked control animals. Despite equivalent stimulation, however, yoked R2>Chr flies did not show increased sucrose consumption relative to yoked controls, arguing against the idea that R2 activation induces a hunger phenotype, which is consistent with our feeding data described above.

We next sought to evaluate the behavioral consequences of R4d inhibition. Interpretation of these experiments, however, was again complicated by the green light required for GtACR1 activation. In our assay conditions, abrupt green-light delivery appeared to be intrinsically aversive or startling, producing reduced interactions and shorter event durations in both R4d>GtACR1 flies and genetic controls. Thus, although trigger flies expressing GtACR1 in R4d neurons displayed aversion to the sucrose well, the similar response observed in trigger flies of the control genotype precluded a clear assignment of valence to R4d inhibition itself (Fig. S2C). Thus, under these conditions, the FLIC assay could not distinguish the effects of R4d inhibition from the effects of light exposure.

Given these limitations, we instead used activation as an alternative means of probing the behavioral state associated with this population. In light-naïve flies, R4d activation produced a behavioral signature closely resembling that of R2 activation: trigger flies exhibited fewer interactions with the sucrose well (Fig. 3C, left $p < 0.01$), shorter visit durations (Fig. S2D, left), and no reduction in the number of well visits (Fig. S2D, right). Thus, as with R2 stimulation, R4d activation appeared to promote rapid disengagement from the trigger well rather than suppressing repeated approach to the sucrose source itself. This aversive effect persisted after one week of pre-treatment (Fig. 3C, center) but was no longer evident after one month of chronic light exposure (Fig. 3C, right) suggesting that, unlike the response to R2 activation, the negative valence associated with R4d activation adapts over time.

We next asked how flies respond when R2 and R4 neurons are activated simultaneously. Based on the aversive effects of activating either R2 or R4d alone, one might expect combined activation to produce a similar or even stronger negative response. Instead, R2/R4 activation increased both the number of interactions with the sucrose well (Fig. 3D, left) and the frequency of visits (Fig. S2E, left), indicating that stimulation reinforced engagement with the trigger-associated food source. This increase in interactions persisted after both one week and one month of pre-treatment (Fig. 3D, center and right, respectively). Median visit duration was unchanged across conditions (Fig. S2E, right). Thus, combined R2/R4 activation did not prolong individual feeding bouts, but instead increased the probability of repeated engagement with the sucrose well. Together, these data indicate that simultaneous activation of R2 and R4 neurons produces a state consistent with positive valence, in contrast to the aversive effects observed when either population is activated alone.

EB valence is shaped by context, sex, and circuit composition

To test whether the valence effects observed in the FLIC generalized beyond a feeding context, we assayed place preference in a rectangular multi-lane arena in which one half of each lane was illuminated and the other remained dark. Unlike the closed-loop FLIC, this open-loop design delivered light on a pre-programmed schedule [specify cycle pattern] regardless of fly position, dissociating valence from sucrose contact or response-contingent stimulation. Flies were observed during a 10-min dark acclimation period followed by a 60-min treatment period. Both sexes were tested, and each manipulation was evaluated at two stimulation frequencies (48 Hz and 200 Hz) as a robustness check. what about the different activation durations? naive

The positive valence of R2/R4 co-activation generalized cleanly. Both sexes significantly preferred the illuminated side during treatment, and it was observed at both 48 Hz (Fig. 7D) and 200 Hz (Fig. S3D), indicating that the appetitive state reflects a stable property of the R2/R4-activated circuit rather than an artifact of a particular assay or stimulation regime.

Consistent with the FLIC results, activation of R2 neurons in females produced a clear avoidance of the illuminated side during the treatment period (Fig. 7B, left), indicating that the negative valence associated with R2 activation extends beyond feeding-related behavior. This effect was also observed under a higher-frequency stimulation regime, supporting the robustness of the phenotype across light conditions (Fig. S3A, left). In males, R2 activation produced the opposite preference: males preferred the illuminated side during both treatment and recovery at 48 Hz (Fig. 7B, right), with a weaker effect at 200 Hz (Fig. S3A, right). Thus, the arena revealed a sex-dependent valence for R2 activation.

By contrast, the FLIC findings for R4d manipulation did not generalize robustly. R4d>GtACR1 and control flies showed indistinguishable side biases (females preferring illumination, males avoiding it; Fig. S3B), so although the open-loop design removed the light-startle confound, it did not permit an unambiguous valence assignment. R4d activation, which had been aversive for naïve flies in the FLIC, produced variable behavior in the arena: females showed little avoidance at 48 Hz but avoided illumination at 200 Hz (Fig. 7C, left; Fig. S3C, left), while males avoided illumination at 48 Hz but showed no preference at 200 Hz (Fig. 7C, right; Fig. S3C, right). The absence of a stable signature across conditions suggests that R4d neurons act in a more context-sensitive or modulatory fashion than R2 or R2/R4.

In summary, the behavioral consequences of EB manipulation are nonlinear and depend on circuit composition, sex, and assay context. R2/R4 co-activation produced a robust positive valence that generalized across feeding-dependent and feeding-independent paradigms and persisted after chronic stimulation, indicating a stable property of circuit state rather than an acute stimulation artifact; in the FLIC, this state was expressed as more frequent returns to the sucrose well without prolonged visit duration, consistent with enhanced approach or incentive salience rather than increased consummatory behavior *per se*. R2 activation was consistently aversive in females across both assays but produced the opposite preference in males, revealing a sex-dependent valence. R4d manipulations, by contrast, did not yield a valence signature stable enough to generalize across assays. Together, these findings argue that EB ring neurons do not make fixed, additive contributions to motivational state, but instead generate qualitatively distinct and differentially stable internal states through combinatorial recruitment, raising the question of how such state differences relate to the common downstream outcome of extended lifespan.

R2 neuron activation remodels metabolism in peripheral tissues

Finally, to examine how EB manipulation globally influences physiology, we performed bulk RNA-sequencing and metabolomic profiling on whole bodies of female flies. Our preliminary investigation was focused on R2-activated flies, as we were especially interested in whether the starvation-response program observed in heads would manifest in the periphery. Body donors were reared and collected the same way as head donors. Bodies were separated from heads using a sieve, and five bodies were collected per sample. Using DESeq2 to identify genes with a significant genotype (CsChrimson, control) and light stimulation (yes, no) interaction, we identified 234 upregulated and 182 downregulated genes in peripheral tissues (experiment-wise p-value < 0.01). Strikingly, the peripheral transcriptional response was nearly completely distinct from that observed in the head, with only 2 of 537 differentially expressed genes shared between compartments (Supplementary Figure XX), indicating that R2 activation drove tissue-specific transcriptional programs. But weren't the bodies exactly those from the heads experiment? If so, you should be clear about that. no; freezer disaster

Despite this disparity in response genes, downregulated processes were consistent with the starvation-like transcriptional profile in heads, as R2 activation drove peripheral downregulation of genes involved in cell cycle progression and DNA synthesis and repair (GO figure). These included replication factors *PolA2*, *Cdc45*, and the Cdt1 ortholog *dup*[31]; mitotic regulators *Cdk1*, *twe* (Cdc25), and *fzy* (Cdc20); kinetochore and centrosome components *Mis12*, *Sas-4* (CENPJ), and *spd-2* (Cep192); and DNA repair factors *RecQ5*, *Mlh1*, and *Pms2*. Chromatin remodeling and transcriptional factors were likewise downregulated, indicating a shift away from proliferative states.

By contrast, we did not observe an upregulation of canonical stress- or starvation-response genes in bodies. Instead, the peripheral response was characterized by electrochemical homeostasis and organellar acidification (cite GO). Among the most significantly upregulated genes were more than a dozen vacuolar H^+-ATPase subunits and accessory factors, including *Vha16-1*, *Vha68-2*, *Vha44*, *VhaAC45*, *VhaSFD*, *Vha26*, *VhaAC39-1*, *Vha55*, *Vha100-2*, and *ATP6AP2*. This V-

ATPase program was not accompanied by the upregulation of autophagy or trafficking pathways, suggesting a decoupling of organellar acidification from degradative or secretory functions. We also noted the upregulation of three inwardly rectifying potassium channels (*Irk1*, *Irk2*, and *Irk3*), possibly to maintain cell membrane potential during increased organellar proton flux. Additionally, we observed enrichment of lipid-associated processes [cite fig], including genes involved in fatty acid synthesis (*ACC*) and ether phospholipid synthesis (*Agps*), suggesting a remodeling of cellular and organellar membranes to support increased ion transport. Notably, we also observed inductions of *Sam-S* and *Gnmt*, indicating changes in methionine flux.

this is a meaty paragraph but I think it is okay at this point. Just finish up with all the data. Targeted metabolomic profiling of peripheral tissues using a similar experimental design, but a less restrictive experiment-wise $p < 0.05$ cutoff, revealed a broad and coordinated metabolic remodeling that reinforeced our interpertation of the transcriptional response. PCA analysis revealed that the single most influential source of variance in the data was the effect of neuronal activation, as PC1 (accounting for 41.2% of the variance) separated R2>Chr flies exposed to light from all other samples(Figure PCA plot). A total of 87 metabolite abundances were found to be significantly affected by R2 neuron activation (Fig heatmap and volcano plots).

Among the 47 metabolites reduced in abundance, the dominant pattern was broad depletion of free nucleotides and nucleosides, including purines, pyrimidines, and modified nucleoside species. A broad reduction in free amino acids and derivatives was also observed. Together, these likely reflect the reduced proliferative burden observed in our transcriptional data. A reduction in S-adenosylhomocysteine (SAH) likely represents accompanying shifts in methylation activity. Reductions in choline, glycerophosphocholine, and acylcarnitines may reflect the previously-described increase in membrane remodeling activity.

The biochemical modulates(?) comprised the 40 signlificantly upregulated metabolites. ()need to establish 3ness) The first of these is a central carbon module comprising glycolytic and TCA-cycle intermediates, including pyruvate, oxaloacetate, and cis-aconitate, alongside pentose phosphate pathway species. This may reflect an accumulation of precursors as a consequence of reduced demand for proliferative inputs. The second module contains small nitrogen species, including glutamine, arginine, serine, and betaine. Glutamine and arginine may likewise accumulate as a consequence of reduced demand from growth-related processes. Serine and betaine serve as methyl donors, and their elevation, along with depletions in SAH and choline and the upregulation of both *Sam-S* and *Gnmt*, indicates a restructuring of methionine cycle flux, potentially recapitulating dietary restriction. Finally, although none of the core kynurenine pathway genes were upregulated in our RNA-seq dataset, we observed a tryptophan-kynurenine module marked by coordinated elevation of kynurenic acid, quinolinic acid, and anthranilic acid, which could account for the depleted tryptophan signature in our data. Pathway-level analysis using FELLA, which maps metabolite changes onto the KEGG network to identify enriched biological modules, corroborated these findings and indicated additional enrichment of taurine and hypotaurine metabolism, beta-alanine metabolism, and folate transport, with starch and sucrose metabolism and nucleotide sugar biosynthesis also represented Fig(FELLA bubble).

3 Discussion

Metab: add something about methionine cycle here. does it phenocopy molecularly? Check lit. Obata and Miura (2015). Move this here: "The convergence of transcriptomic and metabolomic datasets indicates an orchestrated remodeling of peripheral physiology imposed by the activation of a small number of R2 neurons in the brain."

The central finding of this study is that direct manipulation of adult EB ring neurons is sufficient to extend lifespan in *Drosophila*, supporting a model in which the EB acts instructively rather than merely as a passive conduit for lifespan-altering sensory information. Previous work showed that R2 and R4 neurons are required for the lifespan-shortening effects of death perception, establishing these cells as a neural substrate through which sensory experience can influence aging. Here, we show that direct activation of R2 neurons, co-activation of R2/R4 neurons, and inhibition of R4d neurons are each sufficient to promote longevity under standard laboratory conditions. Strikingly, these effects can be elicited by manipulating very small neuronal populations, in some cases fewer than ten neurons per hemisphere, underscoring the sensitivity of organismal aging to defined circuit states. Together, these findings argue that EB ring neurons are not simply necessary for transmitting environmental information relevant to aging, but can themselves generate signals that modulate lifespan.

A second major conclusion is that distinct EB manipulations converge on increased lifespan through divergent proximal molecular states. Bulk head transcriptomics after one week of treatment revealed that R2 activation, R4d inhibition, and R2/R4 co-activation do not produce a common downstream transcriptional program. Instead, R2 activation induced a starvation-like signature that included nutrient-responsive IIS components, autophagy-associated genes, and one-carbon and transsulfuration-associated pathways, whereas R2/R4 co-activation preferentially engaged biosynthetic and storage-associated programs, including amino acid metabolism, glycogen synthesis, and NADPH-generating pathways. R4d inhibition produced a more limited transcriptional response, but attenuated stress-associated changes observed in green-light controls. These results argue against a simple model in which all lifespan-extending EB manipulations act through the same stereotyped physiological state. Rather, they suggest that EB activity can engage multiple downstream programs, each compatible with enhanced longevity.

The transcriptional data also point to specific pathways that may link EB activity to aging-regulatory physiology. The induction of *foxo*, *Thor*, *InR*, *dilp6*, and *Cbs* following R2 activation suggests engagement of nutrient-sensing, proteostatic, and sulfur-amino-acid metabolic processes previously implicated in lifespan control. At the same time, the R2-induced state does not resemble a canonical starvation program. Nutrient-responsive IIS components were induced alongside ribosome biogenesis and translational programs that are not ordinarily co-regulated in this way, suggesting that direct EB activation may uncouple pathways that are typically coordinated by nutrient availability. More broadly, these data are consistent with the possibility that EB activity influences aging by imposing non-canonical nutrient-state programs rather than by simply mimicking physiological starvation or satiety. Because prior work showed that R2/R4 neurons regulate *dilp3* and *dilp5* expression in insulin-producing cells during death perception, one plausible hypothesis is that EB-dependent internal states are translated into systemic aging effects through downstream neuroendocrine centers. However, the present data do not establish that mechanism directly, and defining the relevant effector pathways will require future work.

Notably, the molecular signatures induced by EB manipulation were not mirrored by simple changes in organismal behavior or bulk physiology. Feeding assays provided no evidence that the transcriptional states elicited by R2 or R2/R4 activation were secondary to altered food intake, arguing against dietary restriction or compensatory hunger as the basis of the observed longevity phenotypes. Likewise, changes in TAG stores did not predict starvation resistance, and reduced TAG could occur without increased locomotor activity. Negative geotaxis similarly failed to reveal early behavioral correlates of the transcriptional states measured after one week of manipulation. Together, these results indicate that EB-dependent regulation of aging is not readily explained by gross changes in feeding, energy expenditure, lipid depletion, or locomotor performance. Instead, they support a model in which EB activity imposes internal physiological states whose consequences are reflected more strongly in gene expression than in canonical behavioral or metabolic outputs measured under standard laboratory conditions.

The valence experiments add an important dimension to this model by showing that lifespan-extending EB states need not share the same immediate motivational sign. In the FLIC, R2 activation in females was aversive, promoting rapid disengagement from the sucrose well without suppressing re-approach, while R2/R4 co-activation was positively valenced, increasing interactions and return frequency without prolonging individual feeding bouts. In a feeding-independent place-preference assay, the positive valence of R2/R4 co-activation generalized across sex and stimulation regime, whereas the consequences of R2 or R4d manipulation were more context dependent. These findings indicate that EB ring neurons do not contribute fixed, additive aversive or appetitive signals. Rather, the motivational meaning of EB activity appears to depend on population combination, assay context, sex, and stimulation conditions. Particularly striking is the observation that both aversive and appetitive EB states can promote longevity. This argues against a simple model in which lifespan extension is coupled to one privileged valence state and instead suggests that distinct internal states may engage different downstream mechanisms that converge on slower aging.

This nonlinear relationship between EB population activity and behavioral output may be especially informative in light of known interactions among ring-neuron classes. Prior work suggests that R2, R4m, and R4d neurons participate in an incoherent feedforward circuit, raising the possibility that manipulations such as R2 activation, R2/R4 co-activation, and R4d inhibition may overlap in some network-level consequences even while producing distinct transcriptional and behavioral states. Our data do not resolve whether these interventions ultimately converge on a shared terminal mechanism or instead reach similar lifespan outcomes through partially distinct pathways. However, the marked divergence in both transcriptional state and valence strongly suggests that the relevant aging signal is not encoded by the activity of any single ring-neuron class in isolation. A more parsimonious interpretation is that EB circuits influence aging through population-level configurations whose downstream consequences depend on the particular combination of neurons engaged.

Our findings that discrete populations of ellipsoid body ring neurons can instructively modulate lifespan through the generation of distinct internal states carry potentially significant implications for understanding the neural basis of aging in humans. The *Drosophila* central complex, of which the ellipsoid body is a core component, shares strong neuroanatomical and functional homology with the vertebrate basal ganglia. Both structures serve as integrative hubs that process sensory information, coordinate motor output, and—critically—contribute to the generation of motivational and affective states. Recent work has expanded our understanding of the basal ganglia beyond its classical role in movement control, implicating this structure in the recognition of emotional stimuli, inference of others' emotional states, and awareness of subjective well-being Gendron, 2023 128. In particular, limbic basal ganglia output pathways, including ventral pallidal circuits, can assign appetitive or aversive value to ongoing behavior and reshape responses to challenge in a state-dependent manner. Our findings raise the possibility that a similarly general principle applies to aging: circuit configurations that impose distinct motivational or physiological states may converge on long-term effects on organismal maintenance and survival. More generally, these results suggest that lifespan may be modulated not only by what animals perceive, but by how neural state transforms the meaning of those perceptions into coordinated physiological output. If analogous circuits in the human basal ganglia similarly encode valence states that influence systemic physiology, this would provide a mechanistic framework for understanding associations between psychological states, chronic stress, and aging in human populations.

Several limitations should be considered when interpreting these findings. First, our transcriptomic analyses were performed on female heads at a single early time point, and thus do not capture male-specific or later-emerging molecular states. This is especially relevant given the sex-specific behavioral effects observed in place preference and negative geotaxis. Second, bulk head RNA-seq cannot distinguish transcriptional changes arising within the brain from those occurring in peripheral head tissues, nor can it resolve cell-type-specific responses downstream of EB manipulation. Third, the green light required for GtACR1 complicated behavioral interpretation of some inhibition experiments, particularly in the FLIC, where abrupt light onset itself appeared to be startling. Finally, although the present work shows that EB manipulations are sufficient to alter lifespan and to impose distinct transcriptional and motivational states, it does not yet identify the downstream endocrine, metabolic, or peripheral effectors that execute these aging phenotypes. These limitations define clear priorities for future work, including cell-type-resolved transcriptional analysis, direct study of IPC and neuroendocrine outputs, and more refined dissection of sex-specific responses.

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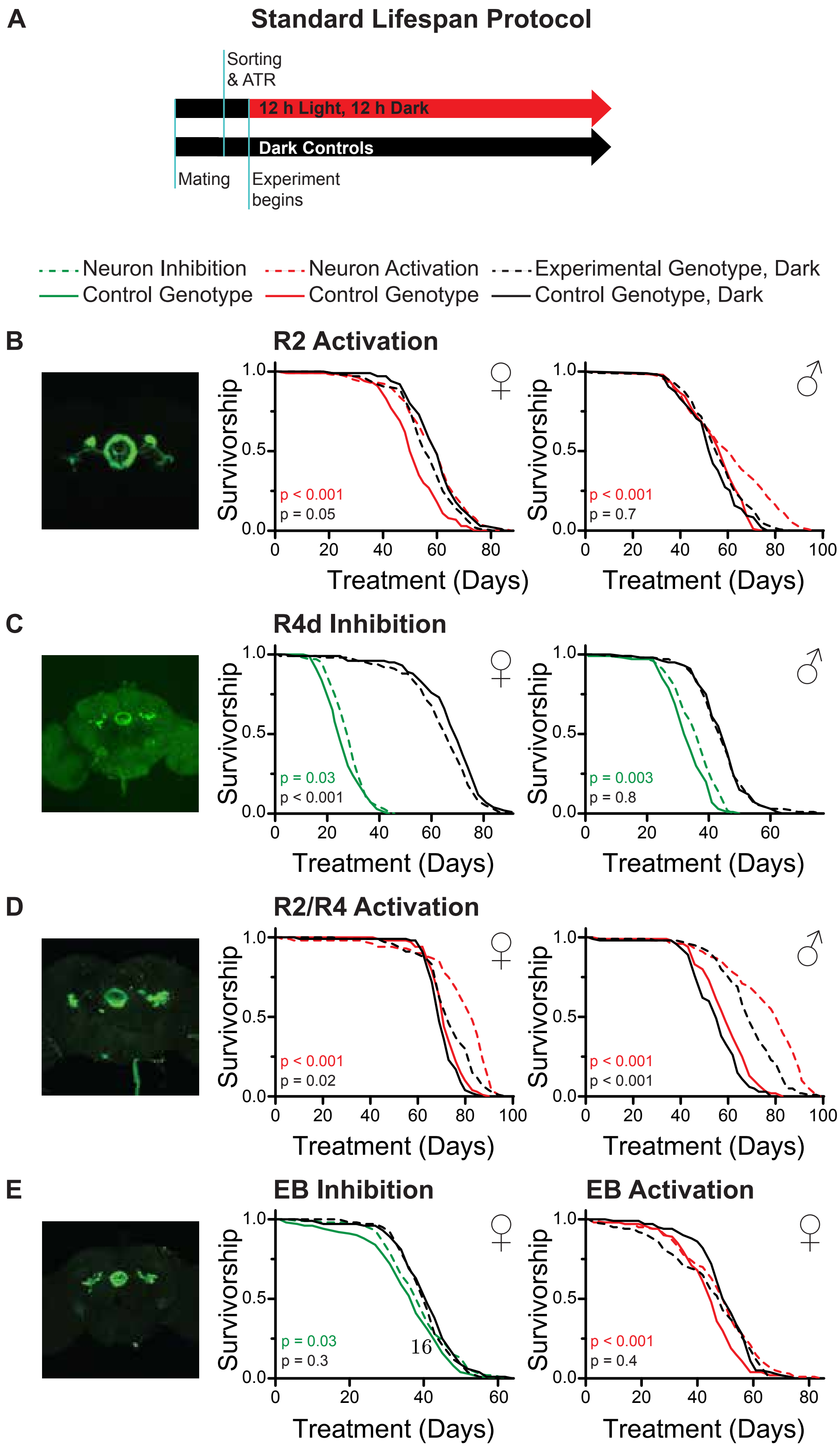
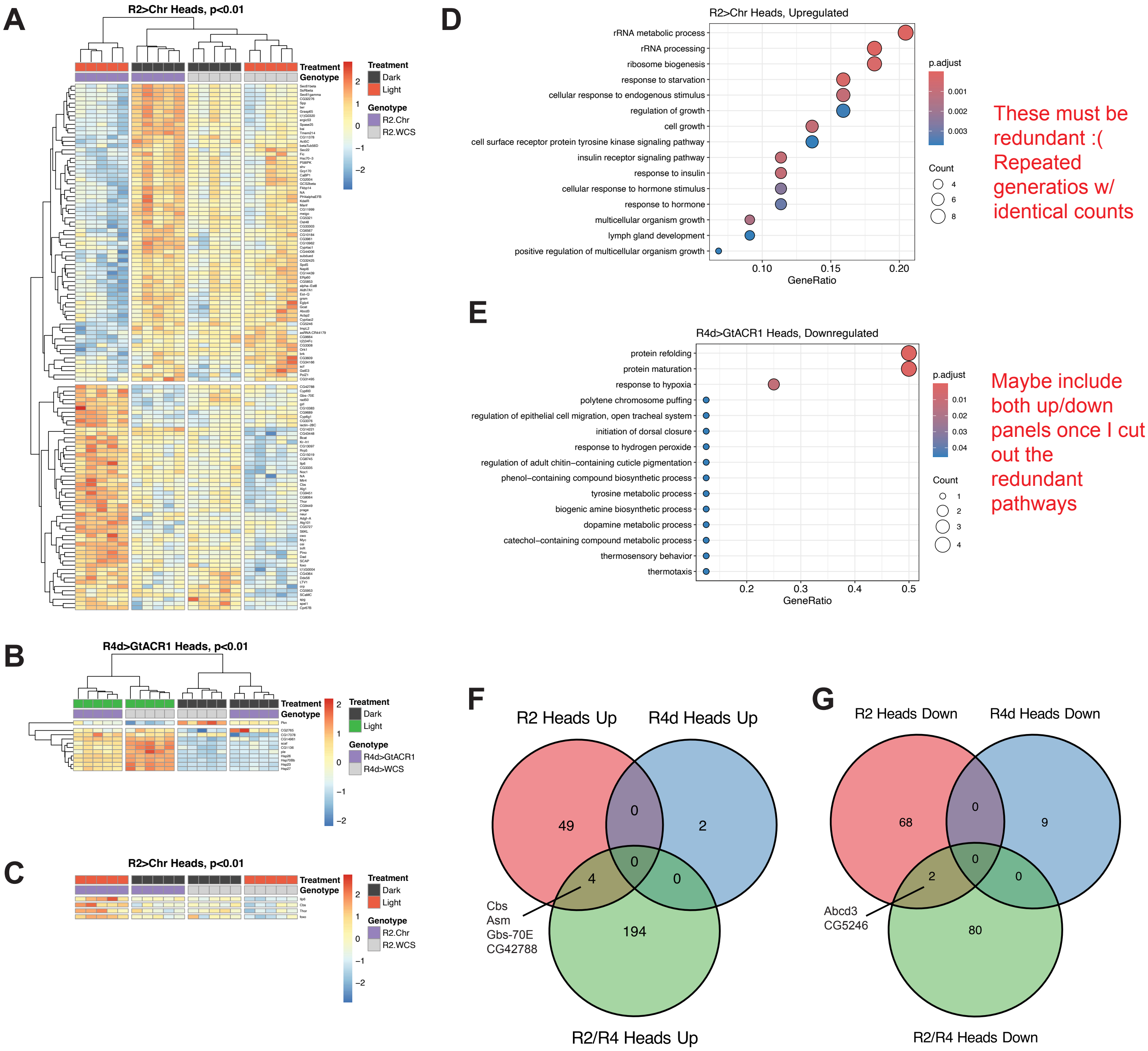
Figures

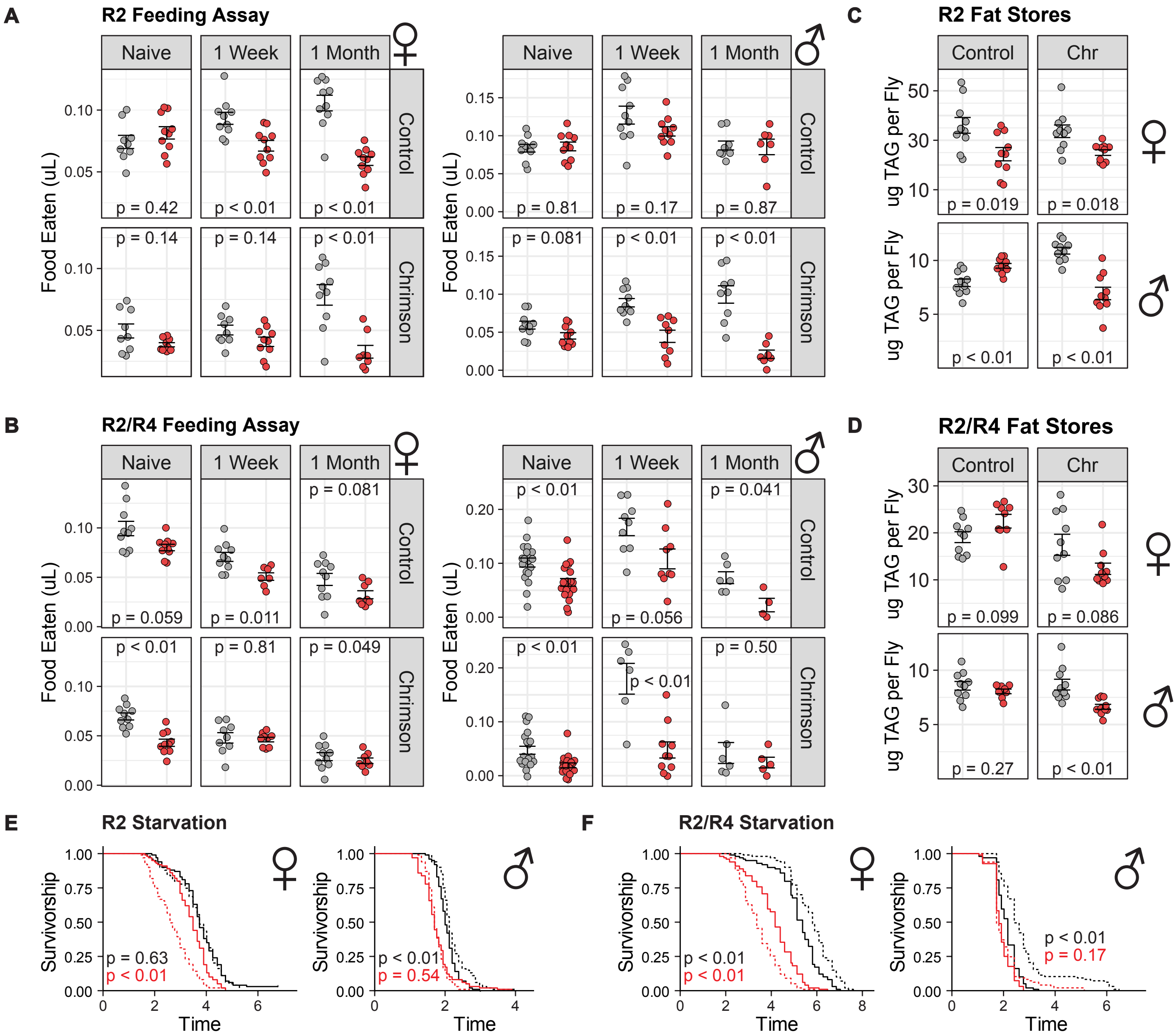


A
Standard Lifespan Protocol
Sorting & ATR
12 h Light, 12 h Dark
Dark Controls
Mating
Experiment begins
Neuron Inhibition
Control Genotype
Neuron Activation
Control Genotype
Experimental Genotype, Dark
Control Genotype, Dark
B
R2 Activation
C
R4d Inhibition
D
R2/R4 Activation
E
EB Inhibition
EB Activation
Survivorship
Treatment (Days)
♀
♂
p < 0.001
p = 0.05
p < 0.001
p = 0.7
p = 0.03
p < 0.001
p = 0.003
p = 0.8
p < 0.001
p = 0.02
p < 0.001
p < 0.001
p = 0.03
p = 0.3
p < 0.001
p = 0.4



These must be redundant :(Repeated generatios w/ identical counts

Maybe include both up/down panels once I cut out the redundant pathways


A
R2 Feeding Assay
Naive
1 Week
1 Month
Control
Chrimson
Food Eaten (uL)
p = 0.42
p < 0.01
p < 0.01
p = 0.14
p = 0.14
p < 0.01
p = 0.81
p = 0.17
p = 0.87
p = 0.081
p < 0.01
p < 0.01
C
R2 Fat Stores
Control
Chr
ug TAG per Fly
p = 0.019
p = 0.018
p < 0.01
p < 0.01
B
R2/R4 Feeding Assay
Naive
1 Week
1 Month
Control
Chrimson
Food Eaten (uL)
p = 0.059
p = 0.011
p = 0.081
p < 0.01
p = 0.81
p = 0.049
p < 0.01
p = 0.056
p = 0.041
p < 0.01
p < 0.01
p = 0.50
D
R2/R4 Fat Stores
Control
Chr
ug TAG per Fly
p = 0.099
p = 0.086
p = 0.27
p < 0.01
E
R2 Starvation
Survivorship
Time
p = 0.63
p < 0.01
p < 0.01
p = 0.54
F
R2/R4 Starvation
Survivorship
Time
p < 0.01
p < 0.01
p < 0.01
p = 0.17

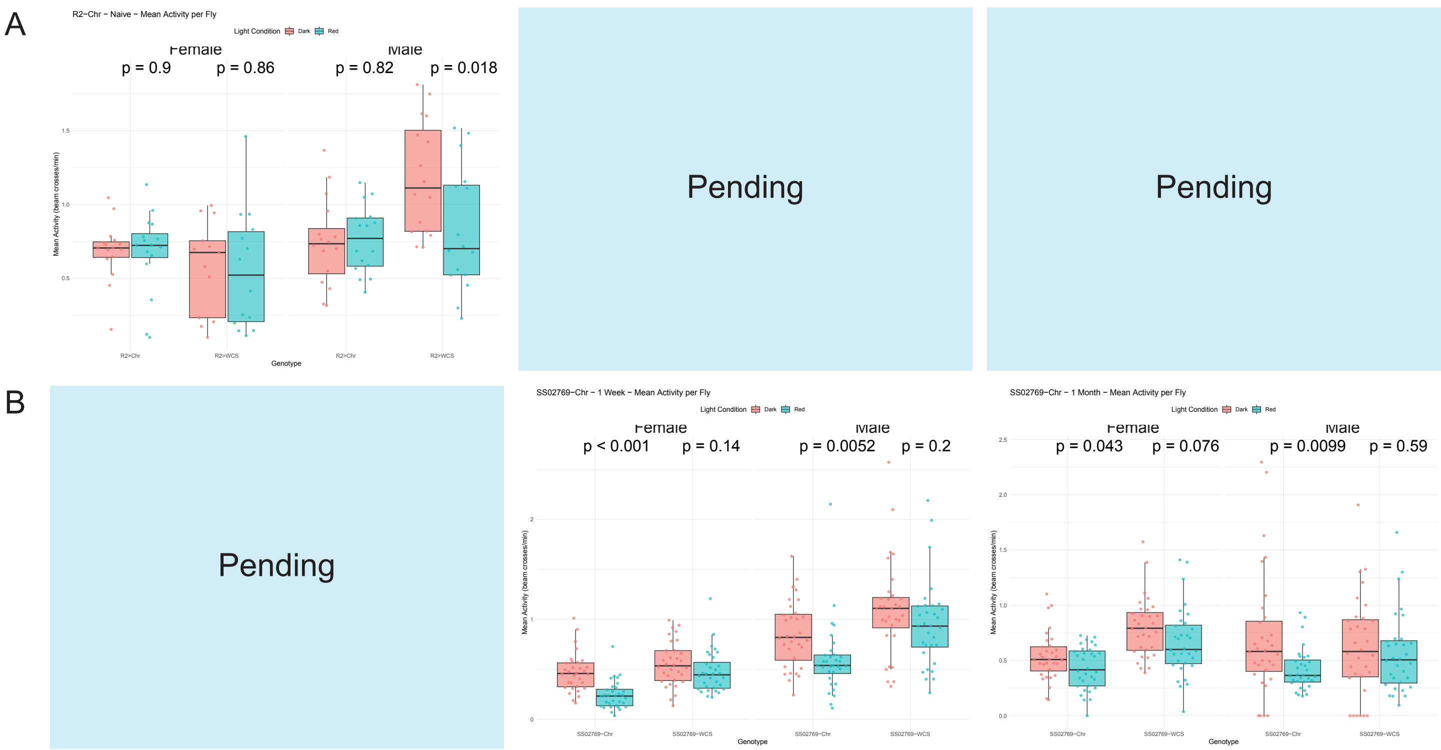
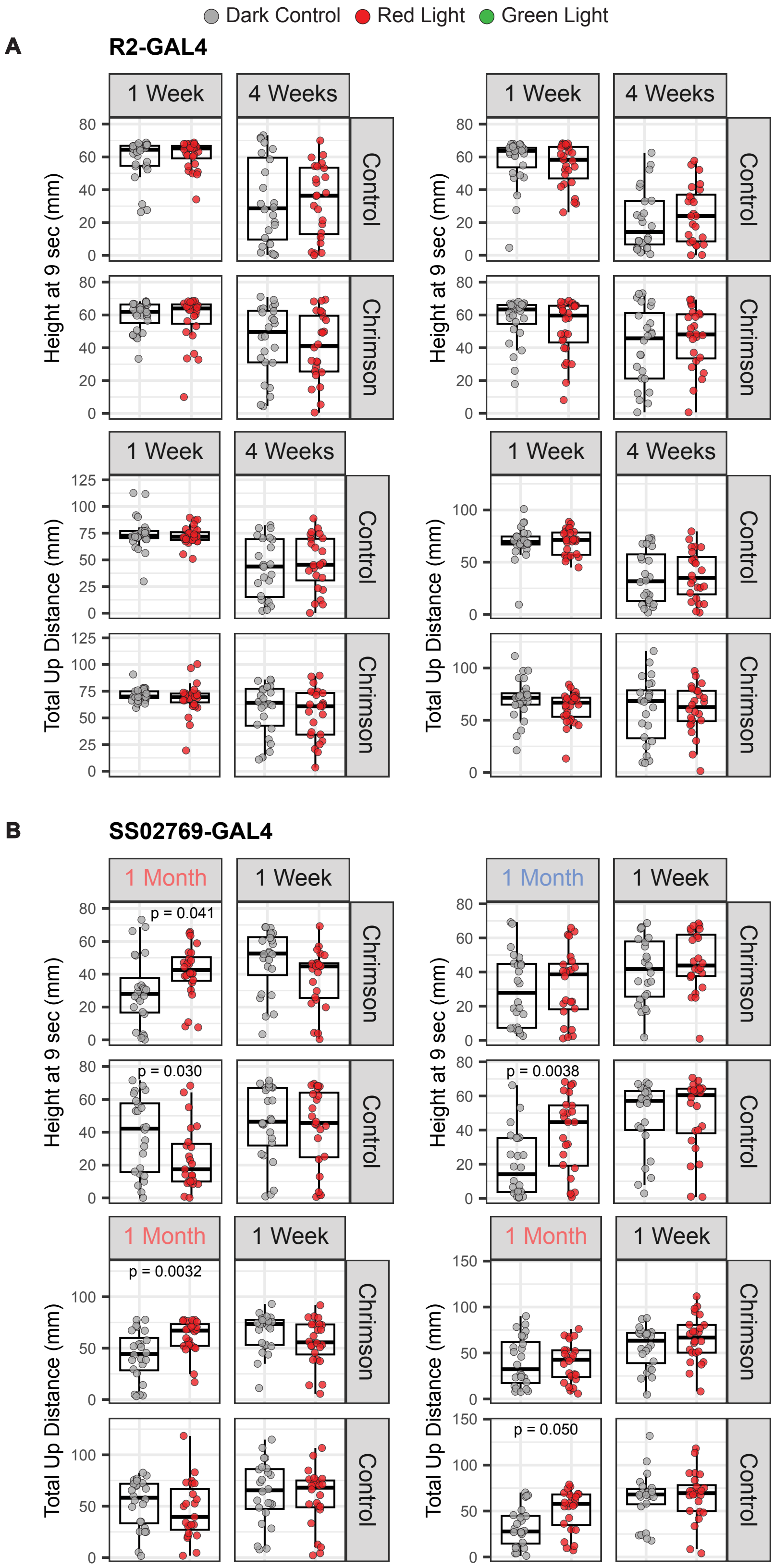
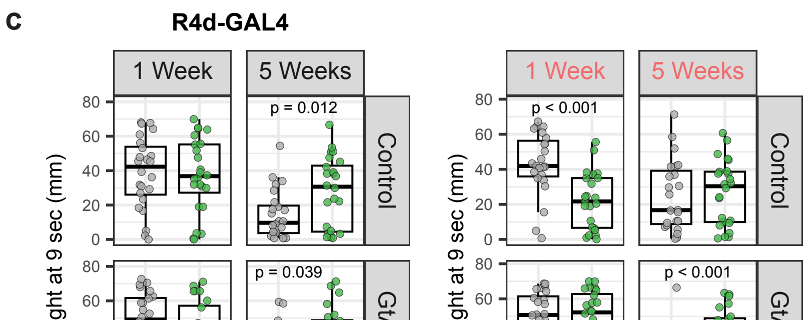


Figure 1. Caption for DAMS Fig. Followed by desc.


Dark Control
Red Light
Green Light
A
R2-GAL4
1 Week
4 Weeks
Control
Chrimson
Height at 9 sec (mm)
Total Up Distance (mm)
B
SS02769-GAL4
1 Month
1 Week
Chrimson
Control
p = 0.041
p = 0.030
p = 0.0038
p = 0.0032
p = 0.050
Height at 9 sec (mm)
Total Up Distance (mm)


C
R4d-GAL4
1 Week
5 Weeks
Control
p = 0.012
p < 0.001
p = 0.039

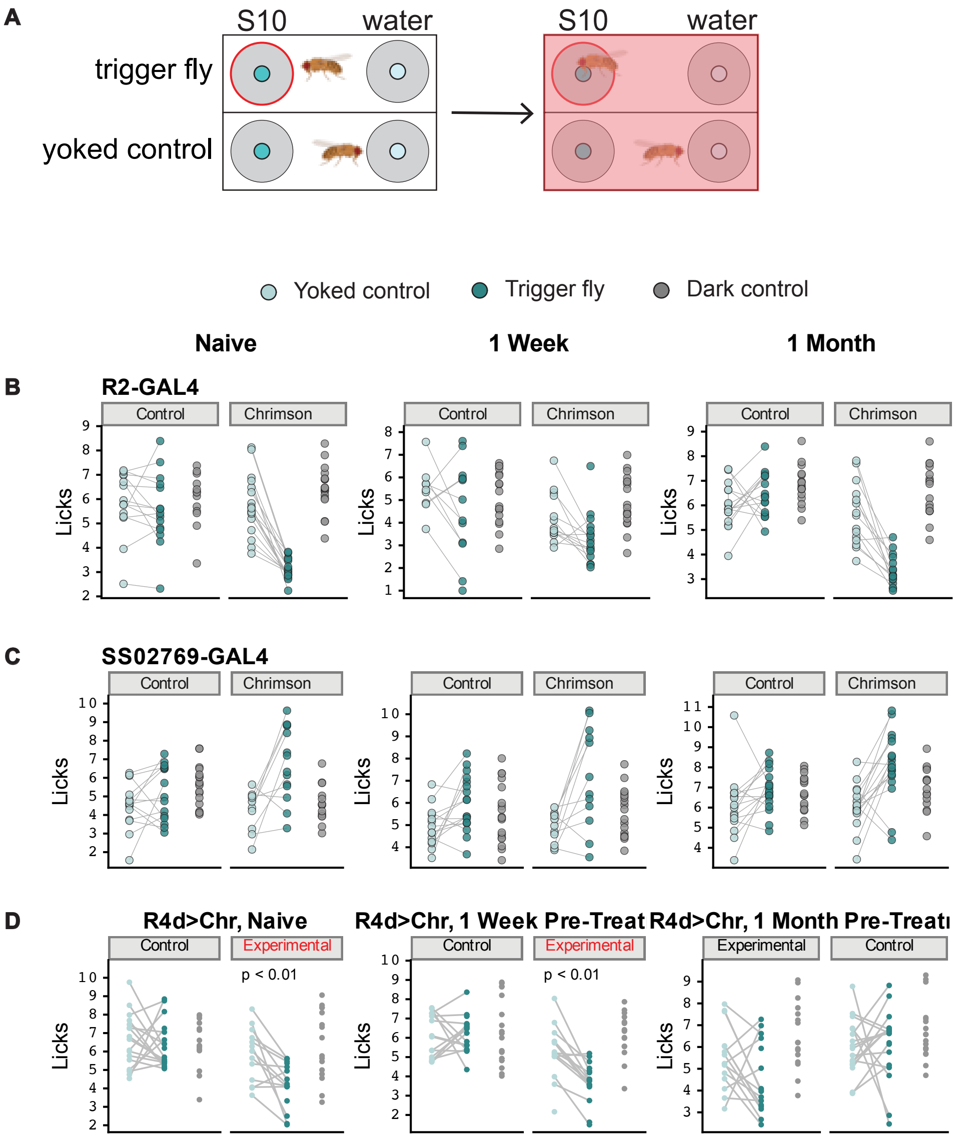


Figure 3. Caption for figure. Followed by desc.

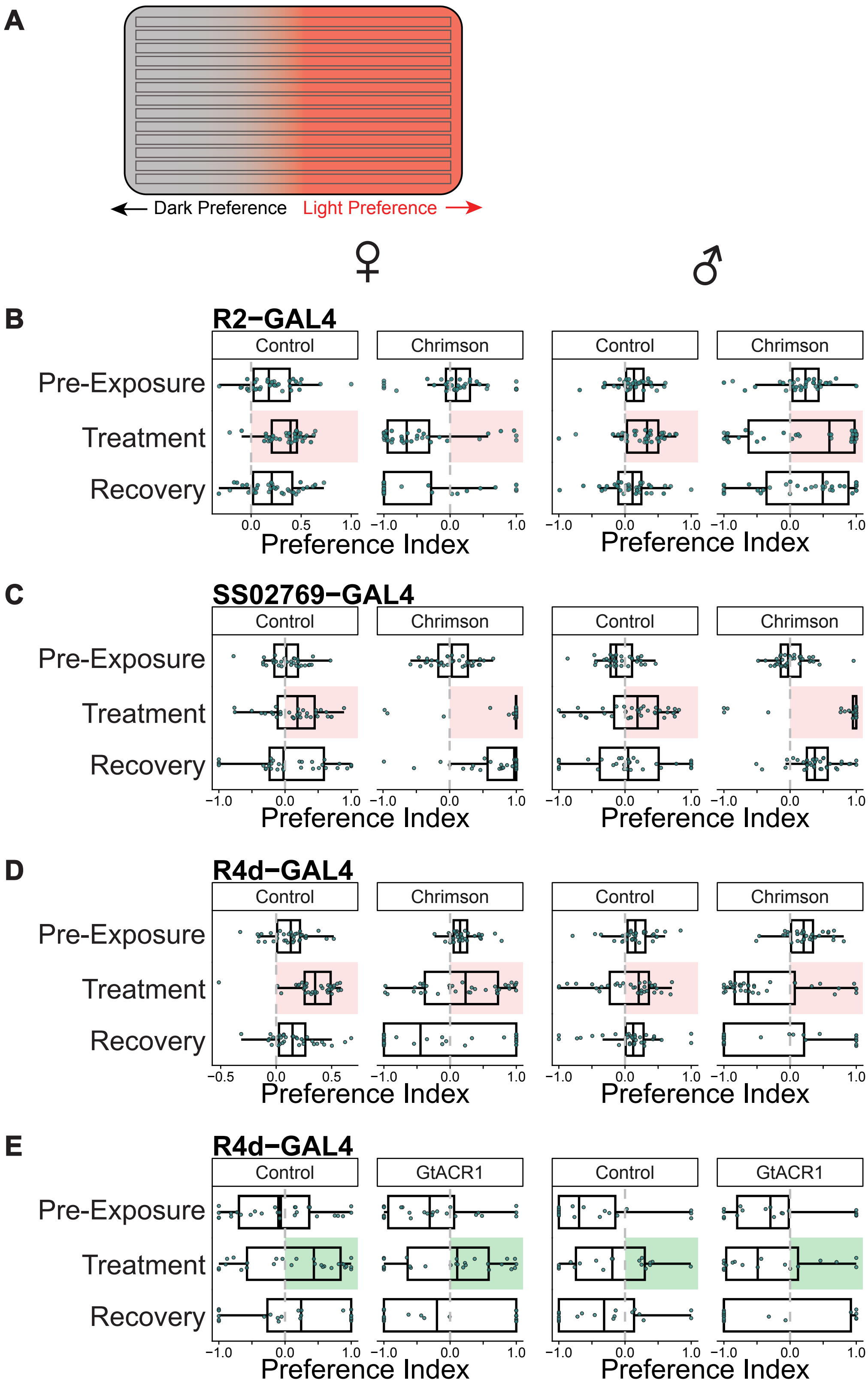

A
Dark Preference
Light Preference
♀
♂
B
R2−GAL4
Control
Chrimson
Control
Chrimson
Pre-Exposure
Treatment
Recovery
Preference Index
Preference Index
C
SS02769−GAL4
Control
Chrimson
Control
Chrimson
Pre-Exposure
Treatment
Recovery
Preference Index
Preference Index
D
R4d−GAL4
Control
Chrimson
Control
Chrimson
Pre-Exposure
Treatment
Recovery
Preference Index
Preference Index
E
R4d−GAL4
Control
GtACR1
Control
GtACR1
Pre-Exposure
Treatment
Recovery
Preference Index
Preference Index

Figure 4. Caption for figure. Manipulation of ellipsoid body neurons R2 and R4 extends lifespan in males and females. (a) Flies were mated for two days, then separated by sex and sorted onto all-trans retinal food for 24 hours before optogenetic treatment began. (b–d) (Left) Expression patterns of drivers used in each experiment. Neurons were either activated (red lines) using UAS-CsChrimson, or inhibited (green lines) using UAS-GtACR1. Dark controls (black lines) were included for all experiments.

Figure 5. Caption for seq. Blah blah blah.

Figure 6. Caption for con-ex. **(A)** R2-Chr. Naive: Light $\times$ Genotype $p = 0.11$ (male) and 0.059 (female). 1-week: Light $\times$ Genotype $p = 0.19$ (male) and $p = 0.15$ (female). 1-month: Light $\times$ Genotype $p < 0.01$ (male) and $p = 0.94$ (female). **(B)** Naive: Light $\times$ Genotype $p = 0.55$ (male) and $p = 0.44$ (female). 1-week: Light $\times$ Genotype $p = 0.064$ (male) and $p = 0.041$ (female). 1-month: Light $\times$ Genotype $p = 0.28$ (male) and $p = 0.23$ (female). **(C)** Light $\times$ Genotype $p < 0.01$ (male) and $p = 0.55$ (female). **(D)** Light $\times$ Genotype $p < 0.01$, males and females. **(E)** R2 starvation stats. **(F)** R2/R4 starvation stats.

Figure 7. Caption for figure. Blah blah arena

Supplemental Figures

R4d>Chr (Supp): - Only diff is 1 week ixn, males and females, for total up distance; exp females worse, ctrl males worse

Lifespan Supp Placeholder

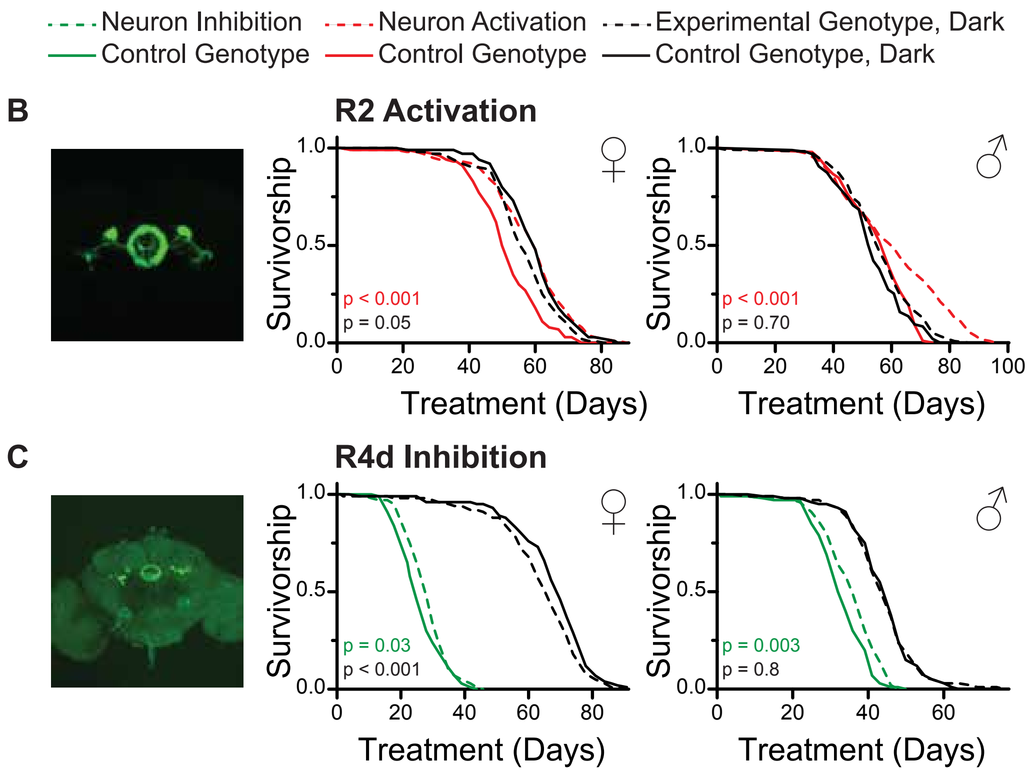


Figure S1. Caption for figure. Followed by desc.

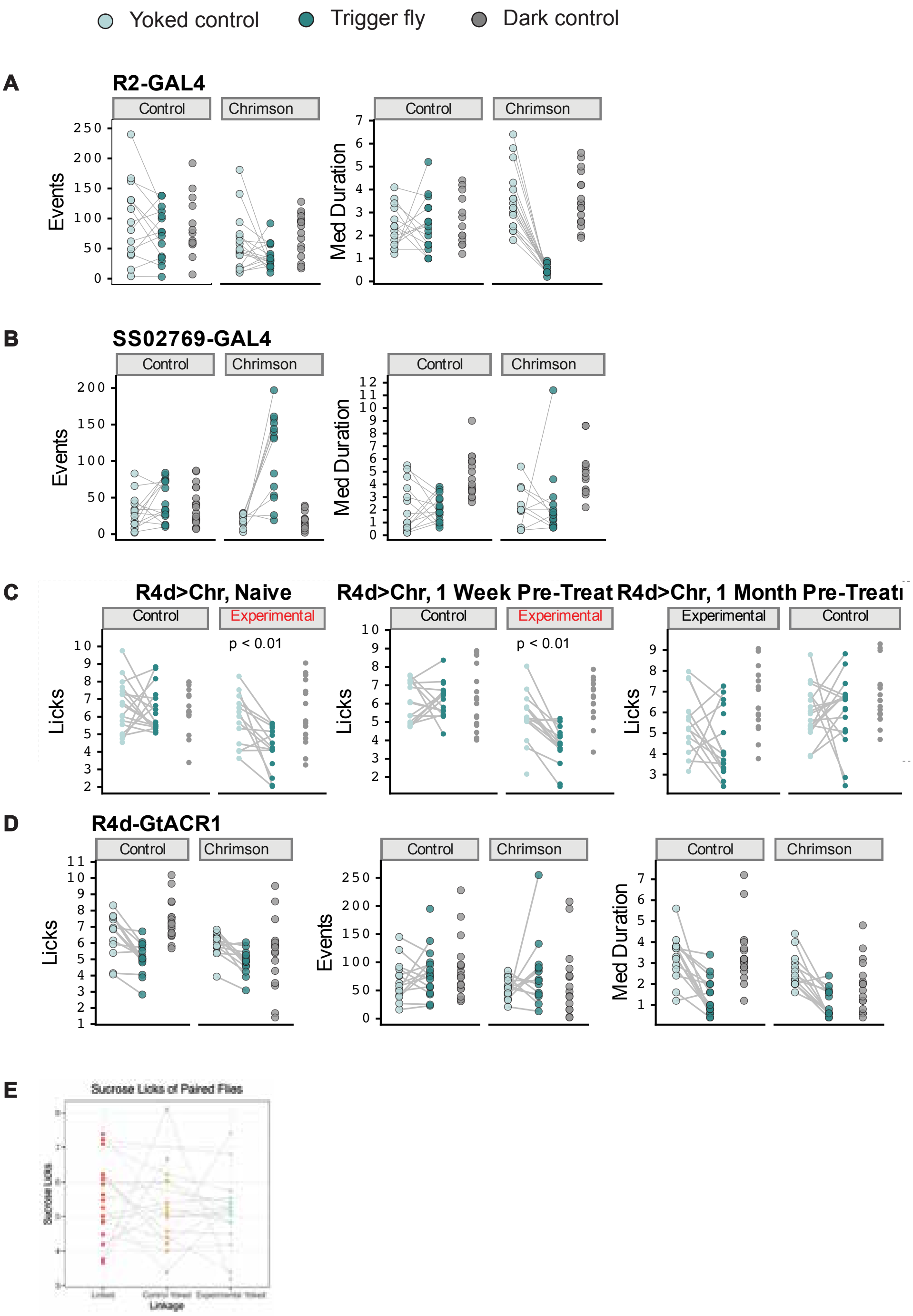


Figure S2. Caption for figure. Followed by desc.

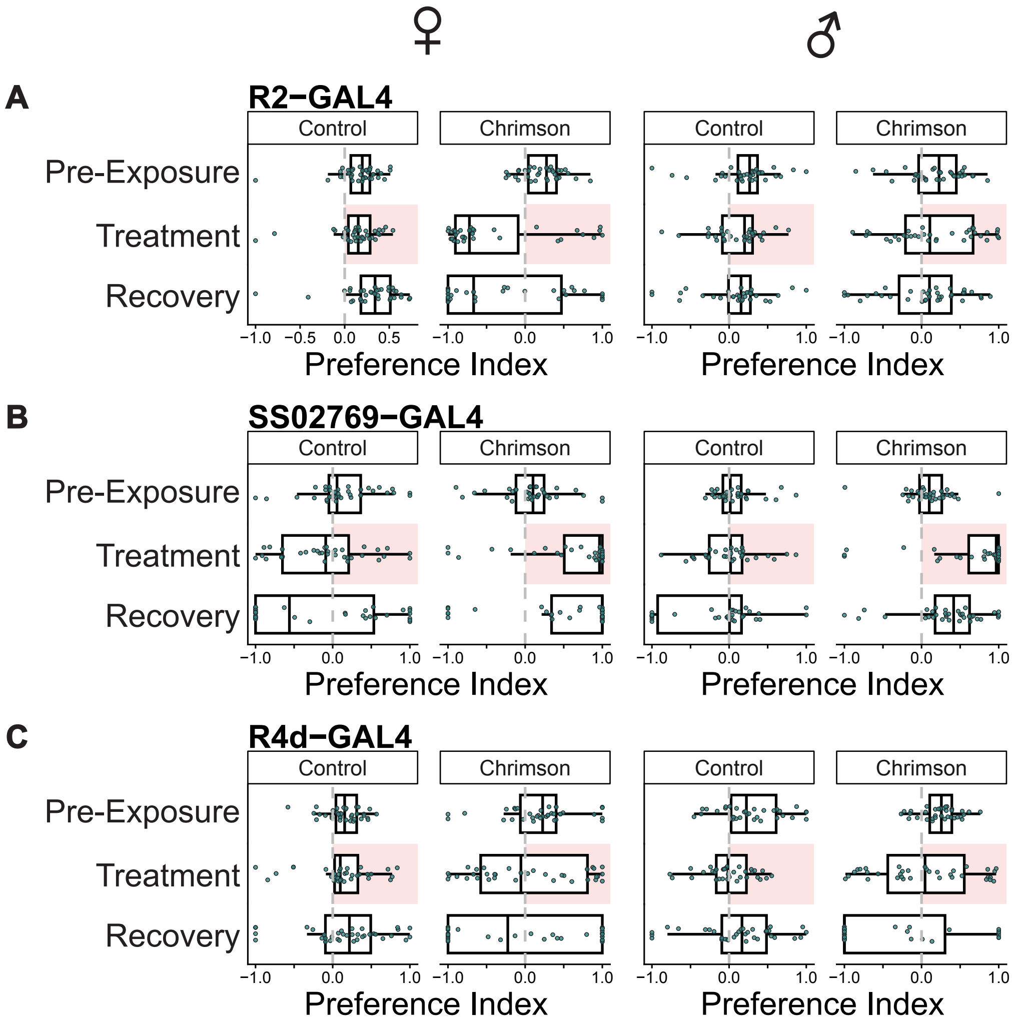


Figure S3. Caption for figure. Followed by desc.

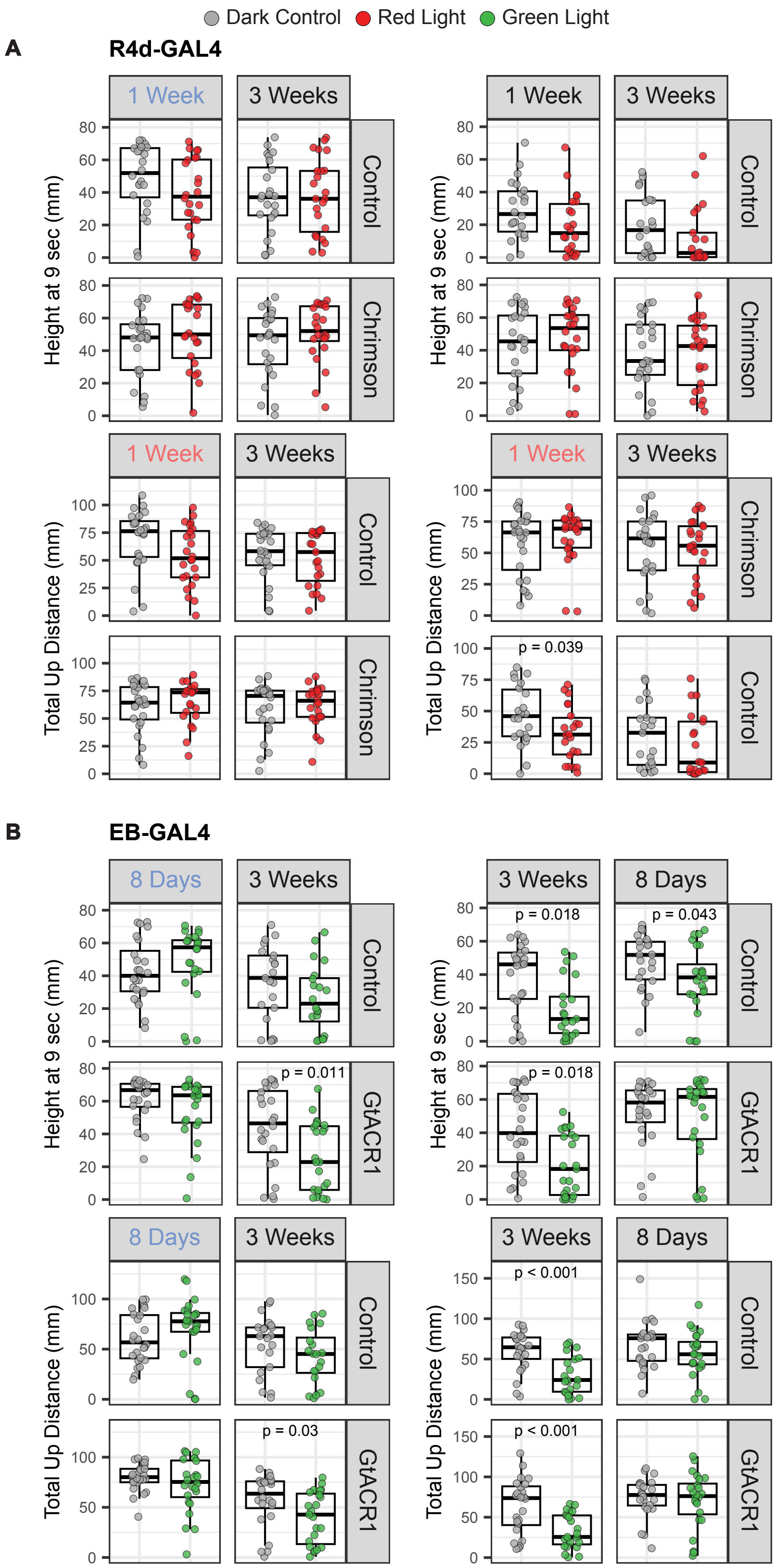


Figure S4. Caption for figure. Followed by desc.